Model-conditioned prior preflight protocol

A lightweight workflow for prior screening, attribution, and recommendation before full training

1 Input

Node-wise trajectory dataset
X_t over N interacting nodes

Candidate prior family
graph / temporal / covariance / other

Model and budget specification
encoder, transition, latent dimension, prior placement, epochs, rollout horizon

Priors are model-conditioned: usefulness depends on the model and training regime.

2 Stage 0: Raw graph-dynamics diagnostics

1) Construct candidate graph / Laplacian (L)

2) Check raw alignment of dynamics with the graph (time)

$D_{\text{true}}(\Delta X)$ or $D_{\text{true}}(\Delta H)$

3) Compute simple graph-dynamics scores

R_{low} | true vs permuted vs random

Fast structural sanity check before training.

3 Stage 1: Quick preflight mini-training

Train lightweight models under the same budget

None | Graph | Permuted graph | Temporal smoothing

$H = 32$ rollout error

- short budget
- same rollout horizon
- same backbone
- same evaluation protocol

Utility can be seen early, but attribution needs controls.

Separate utility from attribution

4 Stage 2: Attribution controls and audits

a) Permuted graph control
Tests whether gain depends on the true topology or only generic smoothing.
effective prior weight

b) Calibrated temporal smoothing
Matches effective regularization strength to test whether graph gain is explainable by temporal smoothing.
effective prior weight

c) Latent-space audit
Check whether rollout gain aligns with graph-smooth latent dynamics.
$D_{\text{true}}(\Delta H)$ | graph vs permuted

5 Outputs

A. Label assignment

- Topology-aligned audit
- Candidate topology-specific
- Temporal sufficient
- Generic graph smoothing
- Low-budget only
- No prior gain

B. Actionable recommendation

- Recommended prior family
- Confidence / caution note
- Whether full training is worth pursuing

The protocol is not a universal truth detector; it is a model-conditioned preflight tool for actionable prior selection.